

An Analysis of Anomaly detection

Anomaly detection

Anomaly detection -- Part 1

The Subspace Outlier Degree (SOD) identifies attributes where a sample is normal, and attributes in which the sample deviates from the expected. Correlation Outlier Probabilities (COP) compute an error vector of how a sample point deviates from an expected location, which can be interpreted as a counterfactual explanation: the sample would be normal if it were moved to that location.

Anomaly detection -- Part 2

Preprocessing Preprocessing data to remove anomalies can be an important step in data analysis, and is done for a number of reasons. Statistics such as the mean and standard deviation are more accurate after the removal of anomalies, and the visualisation of data can also be improved. In supervised learning, removing the anomalous data from the dataset often results in a statistically significant increase in accuracy.

Anomaly detection -- Part 3

Software ELKI is an open-source Java data mining toolkit that contains several anomaly detection algorithms, as well as index acceleration for them. PyOD is an open-source Python library developed specifically for anomaly detection. scikit-learn is an open-source Python library that contains some algorithms for unsupervised anomaly detection. Wolfram Mathematica provides functionality for unsupervised anomaly detection across multiple data types

Anomaly detection -- Part 4

IoT systems Anomaly detection is critical for the security and efficiency of Internet of Things (IoT) systems. It helps in identifying system failures and security breaches in complex networks of IoT devices. The methods must manage real-time data, diverse device types, and scale effectively. Garg et al. have introduced a multi-stage anomaly detection framework that improves upon traditional methods by incorporating spatial clustering, density-based clustering, and locality-sensitive hashing. This tailored approach is designed to better handle the vast and varied nature of IoT data, thereby enhancing security and operational reliability in smart infrastructure and industrial IoT systems.

Anomaly detection -- Part 5

Definition Many attempts have been made in the statistical and computer science communities to define an anomaly. The most prevalent ones include the following, and can be categorised into three groups: those that are ambiguous, those that are specific to

a method with pre-defined thresholds usually chosen empirically, and those that are formally defined:

Anomaly detection -- Part 6

In data analysis, anomaly detection (also referred to as outlier detection and sometimes as novelty detection) is generally understood to be the identification of rare items, events or observations which deviate significantly from the majority of the data and do not conform to a well defined notion of normal behavior. Such examples may arouse suspicions of being generated by a different mechanism, or appear inconsistent with the remainder of that set of data. Anomaly detection finds application in many domains including cybersecurity, medicine, machine vision, statistics, neuroscience, law enforcement and financial fraud to name only a few. Anomalies were initially searched for clear rejection or omission from the data to aid statistical analysis, for example to compute the mean or standard deviation. They were also removed to better predictions from models such as linear regression, and more recently their removal aids the performance of machine learning algorithms. However, in many applications anomalies themselves are of interest and are the observations most desirous in the entire data set, which need to be identified and separated from noise or irrelevant outliers. Three broad categories of anomaly detection techniques exist. Supervised anomaly detection techniques require a data set that has been labeled as "normal" and "abnormal" and involves training a classifier. However, this approach is rarely used in anomaly detection due to the general unavailability of labelled data and the inherent unbalanced nature of the classes. Semi-supervised anomaly detection techniques assume that some portion of the data is labelled. This may be any combination of the normal or anomalous data, but more often than not, the techniques construct a model representing normal behavior from a given normal training data set, and then test the likelihood of a test instance to be generated by the model. Unsupervised anomaly detection techniques assume the data is unlabelled and are by far the most commonly used due to their wider and relevant application.

Anomaly detection -- Part 7

Oil and gas pipeline monitoring In the oil and gas sector, anomaly detection is not just crucial for maintenance and safety, but also for environmental protection. Aljameel et al. propose an advanced machine learning-based model for detecting minor leaks in oil and gas pipelines, a task traditional methods may miss.

Anomaly detection -- Part 8

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Anomaly detection in dynamic networks Dynamic networks, such as those representing financial systems, social media interactions, and transportation infrastructure, are subject to constant change, making anomaly detection within them a complex task. Unlike static graphs, dynamic networks reflect evolving relationships and states, requiring adaptive techniques for anomaly detection.

Anomaly detection -- Part 9

Video surveillance Anomaly detection has become increasingly vital in video surveillance to enhance security and safety. With the advent of deep learning technologies, methods using Convolutional Neural Networks (CNNs) and Simple Recurrent Units (SRUs) have shown significant promise in identifying unusual activities or behaviors in video data. These models can process and analyze extensive video feeds in real-time, recognizing patterns that deviate from the norm, which may indicate potential security threats or safety violations. An important aspect for video surveillance is the development of scalable real-time frameworks. Such pipelines are required for processing multiple video streams with low computational resources.

Anomaly detection -- Part 10

Datasets Anomaly detection benchmark data repository with carefully chosen data sets of the Ludwig-Maximilians-Universität München; Mirror Archived 2022-03-31 at the Wayback Machine at University of São Paulo. ODDS – ODDS: A large collection of publicly available outlier detection datasets with ground truth in different domains. Unsupervised Anomaly Detection Benchmark at Harvard Dataverse: Datasets for Unsupervised Anomaly Detection with ground truth. KMASH Data Repository at Research Data Australia having more than 12,000 anomaly detection datasets with ground truth.

Anomaly detection -- Part 11

Intrusion detection The concept of intrusion detection, a critical component of anomaly detection, has evolved significantly over time. Initially, it was a manual process where system administrators would monitor for unusual activities, such as a vacationing user's account being accessed or unexpected printer activity. This approach was not scalable and was soon superseded by the analysis of audit logs and system logs for signs of malicious behavior. By the late 1970s and early 1980s, the analysis of these logs was primarily used retrospectively to investigate incidents, as the volume of data made it impractical for real-time monitoring. The affordability of digital storage

eventually led to audit logs being analyzed online, with specialized programs being developed to sift through the data. These programs, however, were typically run during off-peak hours due to their computational intensity. The 1990s brought the advent of real-time intrusion detection systems capable of analyzing audit data as it was generated, allowing for immediate detection of and response to attacks. This marked a significant shift towards proactive intrusion detection. As the field has continued to develop, the focus has shifted to creating solutions that can be efficiently implemented across large and complex network environments, adapting to the ever-growing variety of security threats and the dynamic nature of modern computing infrastructures.